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Experiment: Azure Data Factory – Copy Data Pipeline using Azure Blob Storage

1. Resource Group Creation

Create a Resource Group named RG-TANMAY to manage Azure resources.

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The left sidebar shows the 'Resource Manager' section with 'Resource groups' selected. The main content area displays the 'RG-TANMAY' resource group. A list of resources is shown, including 'adf-tanmay01' (Data factory (V2)) and 'storagetanmay01' (Storage account), both located in Central India.

Name	Type	Location
adf-tanmay01	Data factory (V2)	Central India
storagetanmay01	Storage account	Central India

2. Storage Account Creation

Create an Azure Storage Account to store input and output containers.

portal.azure.com/#browse/Microsoft.ClassicStorage%2FStorageAccounts

Uttar Pradesh Sec... Gmail YouTube Maps Police Commissio...

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

1000021031@dit.edu.in
DIT UNIVERSITY (DIT.EDU.IN)

Home

Storage accounts (classic)

☆ ...

Are any resources impacted by outages

Browse all resources in this subscription

Filter resources by region

×

DIT University (dit.edu.in)

+ Create

Manage view

Refresh

Export to CSV

Open query

Assign tags

+ Add to service group

Group by none

Filter for any field...

Subscription equals all

Resource Group equals all

Location equals all

+ Add filter

☐	Name ↑	Type	Kind	Resource Group	Location	Subscription
☐	storagetanmay01	Storage account	StorageV2	RG-TANMAY	Central India	Azure for Students

Showing 1 - 1 of 1. Display count: auto

Give feedback

3. Input Container

Create an Input container and upload the Sample - Superstore.csv file.

portal.azure.com/#view/Microsoft_Azure_Storage/ContainerMenuBlade~/overview/storageAccountId/%2Fsubscriptions%2F98282fc6-afa7-4fcf-92d3-92632a6a5de8...

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Microsoft Azure Search resources, services, and docs (G+)

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Home > Storage accounts (classic) > storagetanmay01 | Containers

input Container

Search

+ Add Directory ↑ Upload 🔒 Change access level ↻ Refresh 🗑 Delete 📄 Copy 📄 Paste 🔄 Rename ⚙ Acquire lease ⚙ Break lease 🛠 Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

input

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

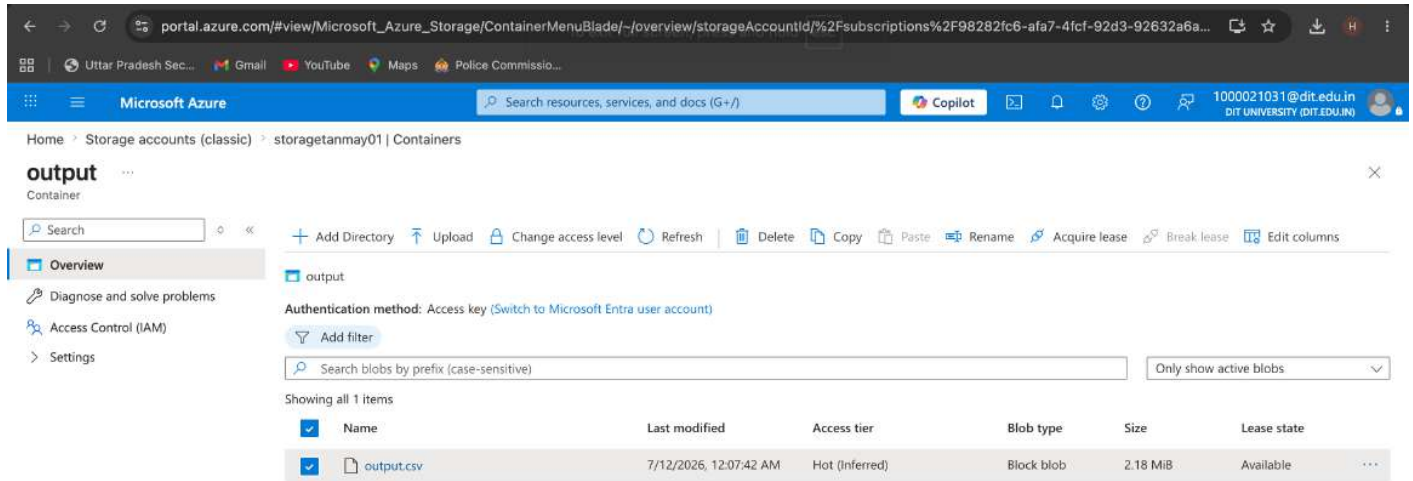
Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Sample - Superstore.csv	7/11/2026, 9:44:01 PM	Hot (Inferred)	Block blob	2.18 MiB	Available

Add or remove favorites by pressing Cmd+Shift+F

4. Output Container

Create an Output container where the copied file will be stored.



The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and a user profile for 1000021031@dit.edu.in. The breadcrumb trail indicates the location: Home > Storage accounts (classic) > storagetanmay01 | Containers. The main content area displays the 'output' container overview. On the left, a sidebar lists navigation options: Overview (selected), Diagnose and solve problems, Access Control (IAM), and Settings. The main area shows the container name 'output' and its authentication method as 'Access key'. Below this, there is a search bar for blobs and a table listing the contents. The table shows one item, 'output.csv', with details on its last modified date, access tier, blob type, size, and lease state.

output

Container

Search

+ Add Directory Upload Change access level Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

output

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

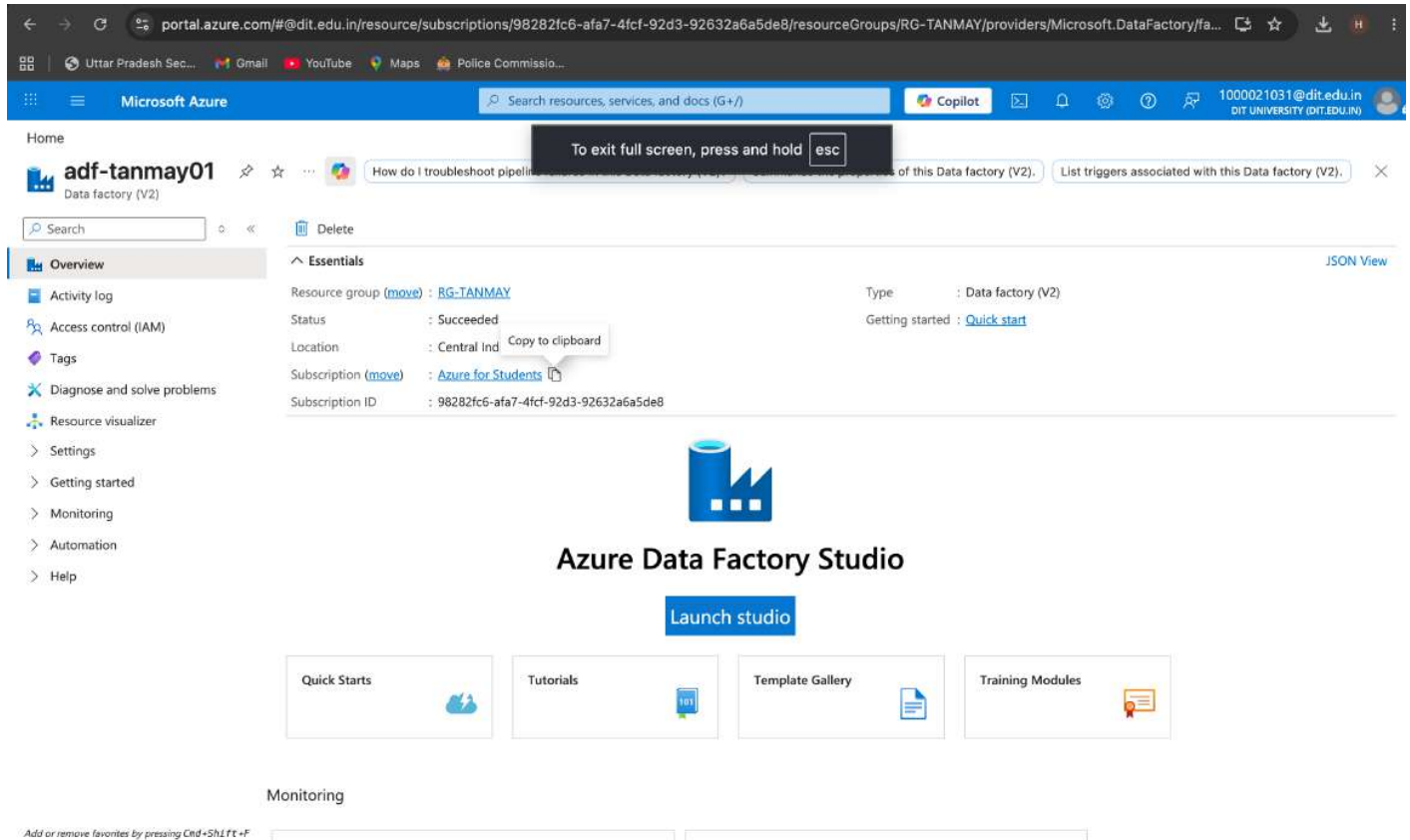
Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Lease state
output.csv	7/12/2026, 12:07:42 AM	Hot (Inferred)	Block blob	2.18 MiB	Available

Add or remove favorites by pressing **Cmd+Shift+F**

5. Azure Data Factory Overview

Azure Data Factory was created successfully and is ready for creating datasets, linked services, and data pipelines.



The screenshot displays the Azure Data Factory Overview page within the Azure portal. The browser address bar shows the URL: `portal.azure.com/#@dit.edu.in/resource/subscriptions/98282fc6-afa7-4fcf-92d3-92632a6a5de8/resourceGroups/RG-TANMAY/providers/Microsoft.DataFactory/factories/adf-tanmay01`. The page header includes the Microsoft Azure logo, a search bar, and the user profile for 1000021031@dit.edu.in. A notification banner at the top states: "To exit full screen, press and hold esc".

The main content area is titled "adf-tanmay01 Data factory (V2)". It features a left-hand navigation pane with the following options: Overview (selected), Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Getting started, Monitoring, Automation, and Help. The "Overview" section is expanded, showing the following details:

- Resource group (move): [RG-TANMAY](#)
- Status: Succeeded
- Location: Central India
- Subscription (move): [Azure for Students](#)
- Subscription ID: 98282fc6-afa7-4fcf-92d3-92632a6a5de8
- Type: Data factory (V2)
- Getting started: [Quick start](#)

A "Copy to clipboard" tooltip is visible over the Location field. Below the details, the "Azure Data Factory Studio" logo is displayed, followed by a "Launch studio" button. Underneath, there are four tiles: "Quick Starts", "Tutorials", "Template Gallery", and "Training Modules". At the bottom, the "Monitoring" section is partially visible, with a note: "Add or remove favorites by pressing Ctrl+Shift+F".

6. Linked Service

Create a Linked Service in Azure Data Factory to connect the Storage Account.

The screenshot displays the Azure Data Factory portal interface. On the left, the navigation pane shows the 'Data Factory' section with 'Linked services' selected. The main area is titled 'Linked services' and shows a table with one item, 'LS_BlobStorage', of type 'Azure Blob Storage'. The right-hand pane is titled 'Edit linked service' and contains the following configuration options:

- Name:** LS_BlobStorage
- Description:** (Empty text box)
- Connect via integration runtime:** AutoResolveIntegrationRuntime (with a green checkmark icon)
- Authentication type:** Account key (dropdown menu)
- Account selection method:** Enter manually (radio button selected)
- Storage account name:** storagetanmay01
- Storage account key:** (Masked with asterisks)
- Partitioned DNS enabled:** (Checkbox, currently unchecked)
- Endpoint suffix:** (Text box)

At the bottom of the 'Edit linked service' pane, there are 'Apply' and 'Cancel' buttons, and a 'Test connection' button with a blue icon.

7. Source Dataset

Create the Source Dataset and configure it to read Sample - Superstore.csv from the Input container.

The screenshot displays the Microsoft Azure Data Factory portal interface. The top navigation bar shows the user is logged in as 1000021031@ditedu.in. The left sidebar lists the 'Factory Resources' including Pipelines (1), Change Data Capture (preview) (0), Datasets (3), Data flows (0), and Power Query (0). Under the 'Datasets' section, 'DelimitedText1', 'DestinationDataset', and 'SourceDataset' are listed. The 'SourceDataset' is selected, and its configuration is shown in the main pane. The configuration is divided into three tabs: 'Connection', 'Schema', and 'Parameters'. The 'Connection' tab is active, showing the following settings:

- Linked service ***: LS_BlobStorage (with links for Test connection, Edit, New, and Learn more)
- File path ***: input / Directory / Sample - Superstore.csv (with a Browse button and a Preview data link)
- Compression type**: No compression
- Column delimiter**: Comma (,)
- Row delimiter**: Default (\r\n, or \r\n)
- Encoding**: Default(UTF-8)
- Quote character**: Double quote (")
- Escape character**: Backslash (\)
- First row as header**: ☒

8. Destination Dataset

Create the Destination Dataset and configure it to write output.csv to the Output container.

The screenshot displays the Microsoft Azure Data Factory portal interface. The top navigation bar shows the URL: `adf.azure.com/en/authoring/dataset/DestinationDataset?factory=%2Fsubscriptions%2F98282fc6-afa7-41cf-92d3-92632a6a5de8%2FresourceGroups%2FRG-TANMA...`. The left sidebar lists 'Factory Resources' including Pipelines (1), Change Data Capture (preview) (0), Datasets (3), and Data flows (0). Under Datasets, 'DelimitedText1' and 'DestinationDataset' are listed. The main panel shows the configuration for 'DestinationDataset' under the 'Connection' tab. The configuration includes:

- Linked service:** LS_BlobStorage
- File path:** output / Directory / output.csv
- Compression type:** No compression
- Column delimiter:** Comma (,)
- Row delimiter:** Default (\r, \n, or \r\n)
- Encoding:** Default(UTF-8)
- Quote character:** Double quote (")
- Escape character:** Backslash (\)
- First row as header:** ☒

9. Copy Pipeline

Create a pipeline with Get Metadata and Copy Data activities.

The screenshot displays the Microsoft Azure Data Factory portal interface for a pipeline named 'CopyPipeline'. The interface is divided into several sections:

- Factory Resources:** A sidebar on the left showing a tree view of resources. Under 'Pipelines', 'CopyPipeline' is selected. Under 'Datasets', 'DelimitedText1', 'DestinationDataset', and 'SourceDataset' are listed.
- Activities:** A central panel showing a search bar and a list of activity categories: Move and transform, Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, General, HDInsight, Iteration & conditionals, Machine Learning, and Power Query.
- Pipeline Design:** The main canvas shows a flow from 'Get Metadata' (containing 'Get Metadata1') to 'Copy data' (containing 'Copy data1'). Both activities have a green checkmark indicating they are configured.
- Settings:** A bottom panel with tabs for Parameters, Variables, Settings, and Output. The 'Settings' tab is active, showing 'Concurrency' set to 1 and 'Elapsed time metric' set to 0.00:10:00.

10. Pipeline Execution

Run the pipeline successfully.

The screenshot displays the Microsoft Azure Data Factory interface. The top navigation bar shows the user is logged in as 1000021031@dit.edu.in. The main content area is divided into three panels:

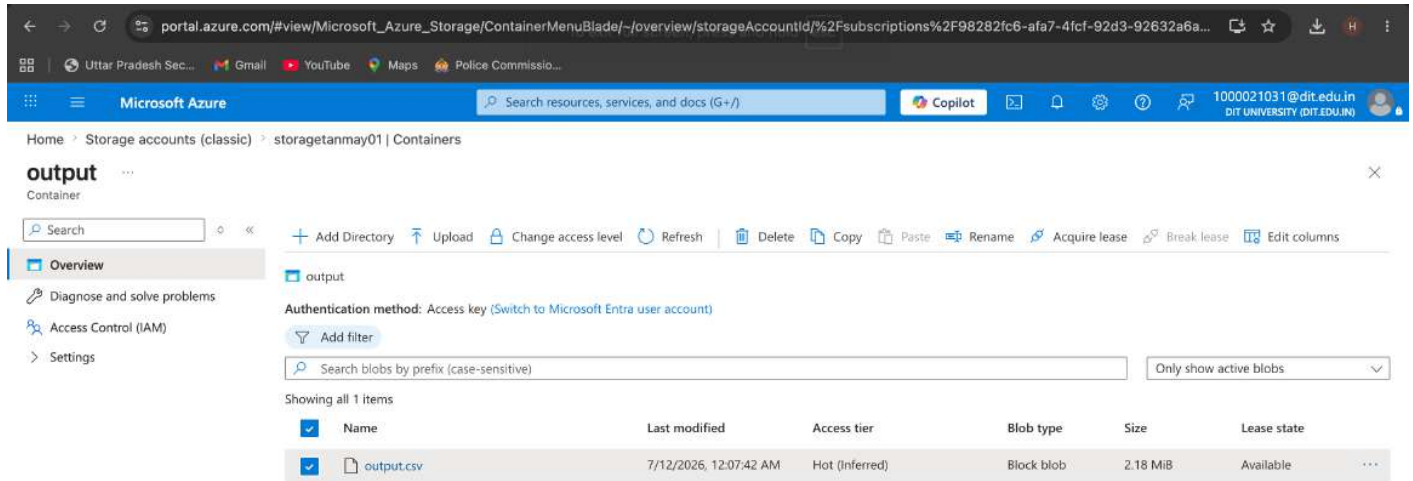
- Factory Resources:** A sidebar on the left showing a list of resources under the 'CopyPipeline' folder, including 'Change Data Capture (preview)', 'Datasets', 'Data flows', and 'Power Query'.
- Activities:** A central panel showing a workflow diagram with two activities: 'Get Metadata1' and 'Copy data1'. Both activities are marked with a green checkmark, indicating successful execution.
- Output:** A table at the bottom showing the execution details of the pipeline run.

The 'Output' table provides the following data:

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	7/12/2026, 5:00:56 PM	16s	AutoResolveIntegrationRu
Get Metadata1	✓ Succeeded	Get Metadata	7/12/2026, 5:00:34 PM	22s	AutoResolveIntegrationRu

11. Output Verification

Verify that output.csv is successfully created in the Output container.



The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and a user profile. The main content area displays the 'output' container. On the left, there is a sidebar with 'Overview', 'Diagnose and solve problems', 'Access Control (IAM)', and 'Settings'. The 'Overview' section is active, showing a table of blobs. The table has columns for Name, Last modified, Access tier, Blob type, Size, and Lease state. One item, 'output.csv', is listed with a last modified date of 7/12/2026, 12:07:42 AM, an access tier of 'Hot (Inferred)', a blob type of 'Block blob', a size of 2.18 MiB, and a lease state of 'Available'.

Name	Last modified	Access tier	Blob type	Size	Lease state
output.csv	7/12/2026, 12:07:42 AM	Hot (Inferred)	Block blob	2.18 MiB	Available

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